Inclusive Rewards as a First Step Towards Evolutionary Multi-Agent Autocurricula

Andries Rosseau
Vrije Universiteit Brussel
Brussels, Belgium
andries.rosseau@vub.be

Raphaël Avalos
Vrije Universiteit Brussel
Brussels, Belgium
raphael.avalos@vub.be

Ann Nowé
Vrije Universiteit Brussel
Brussels, Belgium
ann.nowe@vub.be

ABSTRACT

The competitive and cooperative forces of natural selection have driven the evolution of intelligence for many millions of years, eventually culminating in nature's vast biodiversity and the complexity of our human minds. In this paper, we present a novel multi-agent reinforcement learning framework, inspired by the process of evolution. We assign a genotype to each agent, and propose an *inclusive reward* that optimizes for the fitness of an agent's genes. Since an agent's genetic material can be present in other agents as well, our inclusive reward also takes genetically related individuals into account. We study the effect of inclusion on the resulting social dynamics in two network games, and find that our results follow well-established principles from biology. Furthermore, we lay the foundation for future work in a more open-ended 3D environment, where agents have to ensure the survival of their genes in a natural world with limited resources. We hypothesize the emergence of an arms race of strategies, where each new strategy will be a gradual improvement in response to an earlier adaptation from other agents, effectively creating a multi-agent autocurriculum similar to biological evolution. Our evolutionary autocurriculum provides a novel social dimension that features a non-stationary spectrum of cooperation due to the finite environmental resources and changing population distribution. It has the potential to create increasingly advanced strategies, where agents learn to balance cooperative and competitive incentives in a more complex and dynamic setup than previous works, where agents were often confined to predefined team setups that did not entail the social intricacies that biological evolution has. We argue this could be an important contribution towards creating advanced, general and socially intelligent agents.

KEYWORDS

Multi-agent; Reinforcement Learning; Evolution; Cooperation; Autocurriculum; Networks; Open-Endedness

1 INTRODUCTION

Creating intelligent agents with the ability to adapt to a diverse set of challenges and environments is a prominent goal of artificial intelligence research. In the past decades, the field of single-agent Reinforcement Learning (RL) [40] has made great progress in developing agents capable of completing tasks provided in the form of a reward signal [1, 15, 27, 28, 36]. However, in traditional single-agent RL, once an agent has learned to master its given task, learning stops, since there is no further incentive for improvement. This makes it difficult to create agents which can solve a wide variety of complex tasks; an important characteristic of general intelligence. Creating extra handcrafted tasks for an agent to train and generalize on (i.e., transfer learning [42]) can mitigate this, but this approach is resource intensive and still caps the final complexity that the agent can achieve. Recent work on procedurally generated tasks and environments [10, 38, 46] attempts to solve this problem, but creating adequate reward signals in the environments still remains costly.

Another problem is that an objective can be too complex to learn from scratch. In this case, the space of possible policies is too large to cover effectively with regular exploration strategies. This leads to an agent that cannot close the gap between its initial random behavior and solving the task, getting stuck in low-performing suboptima. Intermediate rewards (such as reaching checkpoints in a maze) can be created to help the agent learn the final overarching goal, but over-engineering the reward signal can lead to problems like specification gaming [2, 21] and potentially limits the range and originality of the learned strategies. A slightly different approach is the use of curriculum learning [5, 11, 29], in which an agent gradually progresses from an easy environment (e.g., a small maze) towards more difficult ones (the full maze), similar to how human education works. Curriculum learning can provide an efficient way of learning complex tasks, but the final complexity is limited by the most advanced task, and creating suitable progressions in the curriculum is resource intensive. Improved exploration strategies can help an agent to avoid getting stuck in suboptimal states. Work in the field of intrinsic rewards proposes ways of overcoming a sparse reward signal. For example, curiosity driven RL [8, 9, 33] provides a self-supervised reward signal which promotes the exploration of previously unknown environment dynamics, often leading to the development of useful skills solely by following the intrinsic reward.

Nature, however, is not a single-agent system, but a multi-agent world full of evolving organisms. The competitive and cooperative forces of natural selection have driven the evolution of intelligence for many millions of years, culminating in nature's great biodiversity and the richness of our human minds. In nature, when a strategy with an increased fitness emerges, it changes the environment dynamics for others, creating a new set of challenges to adapt to [48]. The agents that successfully adapt to these challenges have in turn improved their strategies, thereby again providing new challenges, and so forth. Less successful agents, which are unable to keep up, go extinct. Agents are therefore always at a similar level, and provide just the right amount of challenge for growth. This can be applied to reinforcement learning as well: learning agents continuously improve, thereby pushing the others to adapt, leading to the emergence of a multi-agent *autocurriculum* [23]. Multi-agent autocurricula provide a scalable way for agents to explore a large

---

Proc. of the Adaptive and Learning Agents Workshop (ALA 2022), Cruz, Hayes, da Silva, Santos (eds.), May 9-10, 2022, Online, https://ala2022.github.io/. 2022.

strategy space by simply following the gradients of their experience, called "exploration by exploitation" [4, 23].

In theory, an autocurriculum enables the possibility of unbounded growth for innovation, limited only by the strategy space of the environment and the agents' learning capacities. Autocurricula have formed the backbone for some of the most advanced forms of artificial intelligence known to date. In the form of self-play, it has led to agents with superhuman skill levels in the two-player zero-sum games of Backgammon [43], Go, Chess and Shogi [37], and the continuous real-time strategy game of StarCraft II [44]. In team-based competitive environments, it has led to agents beating the world champions in the real-time strategy game of Dota2 [6], to human-level performance in a first-person 3D multiplayer game of capture-the-flag [20], and to an arms race in a 3D hide-and-seek game [4], where several distinct strategic phases emerged, each requiring increasingly sophisticated forms of cooperation and use of tools.

Our work fits in the tradition of the aforementioned works on multi-agent autocurriculum learning, aiming to create high levels of complexity starting from elegant, simple rules. However, the dynamics in previous work on multi-agent autocurricula are limited to either all competition, or in the case of predefined team setups, all cooperation with competition between teams. Humans – like many other organisms in nature – are not that binary, instead showing a range of cooperative behavior. As the main contribution of this work, we propose the first steps towards a novel multi-agent autocurriculum inspired by biological evolution, where we construct an evolutionary aligned reward based on the survival of an agent's genes. Since other agents potentially carry parts of the same genetic material, the reward function includes the fitness of others as well, weighted by a measure of genetic relatedness. Our reward structure therefore also leads to the emergence of a spectrum of cooperation, based on the relatedness of the genotypes present. This spectrum can shift over time: for example, when the total population size grows, resources become scarce, leading to a tension between helping closer relatives and relatives carrying less of your genetic material. At any given time, the actual levels of cooperation are therefore determined by the population of genotypes, but themselves influence the population of genotypes that will be present in the future. This cycle adds a novel social dimension to the multi-agent autocurriculum, which continuously challenges the agents to find new strategies that balance cooperation and defection appropriately. We argue that our multi-agent autocurriculum has the potential to show a continuous growth in agents' strategic complexity, only bounded by the strategy space of the environment and the agents' learning capabilities.

2 METHODS

2.1 Information stability

We propose a general definition of fitness as the stability of an *information state* with regards to its environment. The more stable an information state is, the longer it will keep existing. For example, galaxies or diamonds are information states with a high fitness in the realm of physics. When considering complex organic molecules on a primeval Earth, the strategy of producing a copy of oneself before being destroyed (i.e., replicating) turned out to be a particularly stable one, drastically lengthening the existence of one's information state. Yet, every so often, an error occurs in the copying process, known as a mutation. When a mutation is beneficial, it leads to an improvement in fitness, which is favored by natural selection. At the same time, however, a mutation also changes the information state. Mutations, coupled with natural selection, generally lead to a gradual shift in the population of replicators, towards increasingly stable (fit) information states. Replicators, which exist today in the form of DNA, have built an astonishing set of ingenious organisms around themselves to help them survive.

In biology, the parts of DNA that code for the observable traits (i.e., the behavior) of an organism are the genes, and together they form a genotype. The genotype represents the complete information state on which selection acts. To translate our definition of fitness into an evolutionary aligned reward function for reinforcement learning, we implement an abstract version of genetics in our agents. We assign an agent i with an abstract genotype $\mathbf{g}_i$, which is a sequence of n genes where each gene locus/index $k \in [1, n]$ contains a gene g_i^k. Different integer values for g_i^k then represent different gene variants, where in principle every gene locus can have an undetermined amount of gene variants .

We propose a metric of information similarity between genotypes to quantify their relatedness. In information theory, the Hamming distance $d(\mathbf{s}_1, \mathbf{s}_2)$ [18] between two sequences $\mathbf{s}_1$ and $\mathbf{s}_2$ is the number of positions at which corresponding entries are different, measuring the amount of substitutions ('bit flips') needed to change one sequence back into the other. Starting from the normalized Hamming distance, we derive a similarity metric, expressing the genetic relatedness between two agents as a real number between 0 and 1, which we name the *Hamming similarity*. Considering two agents i and j, the Hamming similarity is defined as:

$$h(\mathbf{g}_i, \mathbf{g}_j) = 1 - \frac{1}{n} d(\mathbf{g}_i, \mathbf{g}_j) = \frac{1}{n} \sum_{k=1}^{n} \delta(g_i^k, g_j^k), \tag{1}$$

where $\delta(\cdot, \cdot)$ is the Kronecker delta. Note that this metric is defined when both genotypes have the same length. In the case of different genotype lengths, we could use the Damerau-Levenshtein distance [7], an extension of the Hamming distance which takes into account information deletions and insertions as well.

2.2 Inclusion

2.2.1 Inclusive reward function.

Since an agent's genetic material can be present in others as well, helping agents which are genetically related should also be promoted by our reward function. We therefore modify the reward of each agent i by adding the rewards of the other agents as well, multiplied by their Hamming similarity (Eq. 1). We call this modified reward the *inclusive reward*, after the concept of inclusive fitness [16, 34], which posits that under the right circumstances, natural selection favors organisms that help their genetic relatives. We define the inclusive reward r_i^* as:

$$r_i^* = \sum_{j} h(\mathbf{g}_i, \mathbf{g}_j) r_j = r_i + \sum_{j \neq i} h(\mathbf{g}_i, \mathbf{g}_j) r_j \tag{2}$$

An illustrative example of an inclusive reward is given in Figure 1, where we consider two agents playing a prisoner's dilemma [22]. Both agents can either cooperate (C), or defect (D). A rational agent will always choose to defect, since that action always gives more